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JH. String Ensemble / Triple Chorus - "Solina"(TM) Emulator

(Disclaimer: Solina used to be a Trademark of Eminent, and was licenced to ARP when the original String Ensemble keyboard was built. AFAIK the trademark is not active anymore - I searched the US trademark data base and found the status "dead" - but you can never know for sure. I do not intend to infringe anybody's right of exclusive use of their Trademarks. In case someone still *has* the right to the Solina trademark, I kindly ask you to notify me using this [email](#), and I'll remove it from my site immediately.)



Background

The lush Sound of the Solina (TM) Ensemble is created by 3 BBD delay lines that are modulated in a unique way: There are two 3-phase modulation generators, one running at slow speed ("Chorus"), and one running at high speed ("Vibrato").

We'll focus on one of the modulation generators first, "Chorus": the slow one.

"3-Phase" means that the modulation generator has 3 outputs, each of which's phase is roughly 120 deg apart from the previous output.

Let's call them "0 deg", "120 deg" and "240 deg" - it's easy to see that, with 360 deg describing a full circle, the three modulation outputs a modulation generator are equally distributed around a circle. They are routed to the CV inputs of the 3 BBD's clock VCOs. Modulation of a BBD line causes a pitch shift similar to the Doppler effect of a moving sound source, so with the 3 BBD lines modulated by the 3-phase control signals, a sonic image of 3 sound sources that are moving along the same circle, with equal distance to one another along the outline of the circle, is created.

Actually, each BBD clock VCO is controlled not by a single modulation generator, but by a combination of the slow and the fast generator.

BBD1 sees a CV that is combined from the Chorus generator's "0 deg" output and the Vibrato generator's "0 deg" output.

BBD2 sees Chorus "120 deg" and Vibrato "120 deg".

BBD3 sees Chorus "240deg" and Vibrato "240 deg".

This method creates the famous "Solina" sound, which was so sucessfull that it has been emulated by other

manufacturers.

Of these, I have studied two very closely: The Crumar Performer, and the Dr. Boehm Phasing Rotor 78. I've also taken a look at the Korg Polysix's Ensemble mode.

Classic Implementation

The original Solina did not use a 3-phase oscillator at all. Not even a sine wave oscillator, for that matter.

There's a Square Wave oscillator which is then turned into an approximated sine wave by heavy filtering, creating the "0 deg" signal of the "Chorus" part.

This is then fed into a 1-pole low pass filter with a gain 1.83 for very low frequencies. The brilliant idea behind this: For one specific input frequency, you get unity gain *and* the desired 120 degree phase shift. You can adjust the frequency of the square wave generator for the "0 deg" and "120 deg" signals to have the same amplitude, and you'll have the right frequency and the right phase shift automatically.

The "240 deg" signal is not created with another filter: It's derived from the "0 deg" and "120 deg" signal with a simple inverting adder stage. (The wonders of vector maths: The vector sum of a balanced 3-phase system is always zero.)

The "Vibrato" path even needs one stage less: There's the square wave oscillator with filtering for the "0 deg" signal, a LPF for the "120 deg signal".

These two signals are added to the "Chorus" signal chain at the respective stages, so the inverting adder of the "Chorus" takes care of the vector maths to create the "240 deg" signal of the "Vibrato" as well. (Even though there is no point in the circuit where you could measure the individual "240 deg" signals separately!)

This was pretty brilliant at a time when saving an opamp stage in a circuit made a difference. :)

From today's point of view, it's the tiny flaws of the original method that may make it interesting: Whatever adjustment of the Vibrato and Chorus speed potentiometers, it is *not* possible to create a perfect level balance and equal phase distribution for both, the fast and the slow part of the modulation.

My conclusion is that *if* there is a perceivable difference in the sound of a Solina, compared to a Boehm or the Polysix (other than EQ-ing and SNR), this may have to do with these special modulation waveforms.

The Boehm, for instance, has a very direct and precise method of directly creating 3-phase signals.

My Emulation

When emulating the behaviour of some Vintage instrument, my design goal is always to be as precise as possible. But don't get fooled: If part of that vintage sound is due to certain imperfections, all the precision of the emulation must be directed to re-create that specific - once "non-perfect", but now your reference! - solution.

This doesn't mean to just copy the original and to use the same components. While this certainly works in most cases, it's rarely an option if you want to give others the opportunity to build your circuit as well. I *have* a set of TCA 350's in my drawer, but I'd rather go for BBDs that are still available (even when production has long stopped), the TDA1022. And I've omitted a lot of the more bulky components in the Solina's modulation generators, but I've painstakingly taken care to get the same waveforms, with all their "flaws", as in the original. (A circuit simulator is a gift from heaven for that kind of work.)

I've also noticed that a diode in the Solina's BBD Clock VCOs is slightly bending the combined modulation waveform, before it really hits the VCO core. As my VCO implementation is different - CMOS gates like the Boehm, instead of transistors - that kind of waveform bending is also implemented in a different way, to get the same result *without* introducing an extra 21V supply rail. I've made it adjustable with a trimmer, so you can set that

effect to your taste, Solina Way or Dr. Boehm Way.

Filtering in general, and the Crumar Performer in particular

In the Solina, the signal as it comes from the tone generation and string sound shaping, runs thru a second order 12kHz filter before it goes into the BBDs. This acts as an anti-aliasing filter, as well as adding to the sound shaping a bit.

With longer BBD lines (TDA1022 instead of TCA350), the clock frequency can be higher, this allowing for a wider bandwidth without increased aliasing.

So I've implemented a 16kHz anti-aliasing filter, plus a 12kHz filter, to be able to choose a broadband sound vs. a slightly darker sound.

A third option was inspired by the Crumar Performer: It has a very specialized 3-Band EQ that is very effective on string sounds. With my Performer, I get very convincing, yet very different, string sounds with a 0-10-0 setting and with a 10-0-10 setting (low-mid-high).

I've included this wonderful EQ in my emulation.

New: Schematics

[Schematics, Page 1: Signal Path](#), PDF, also shows Mode Selector as 3-Position Switch (Rotary Switch)

[Schematics, Page 2: Modulation](#), PDF

[Connecting Switches and Pots, Option 1](#), PDF, shows Mode Selection as Set of 2-Position Switches (Toggle Switches etc.)

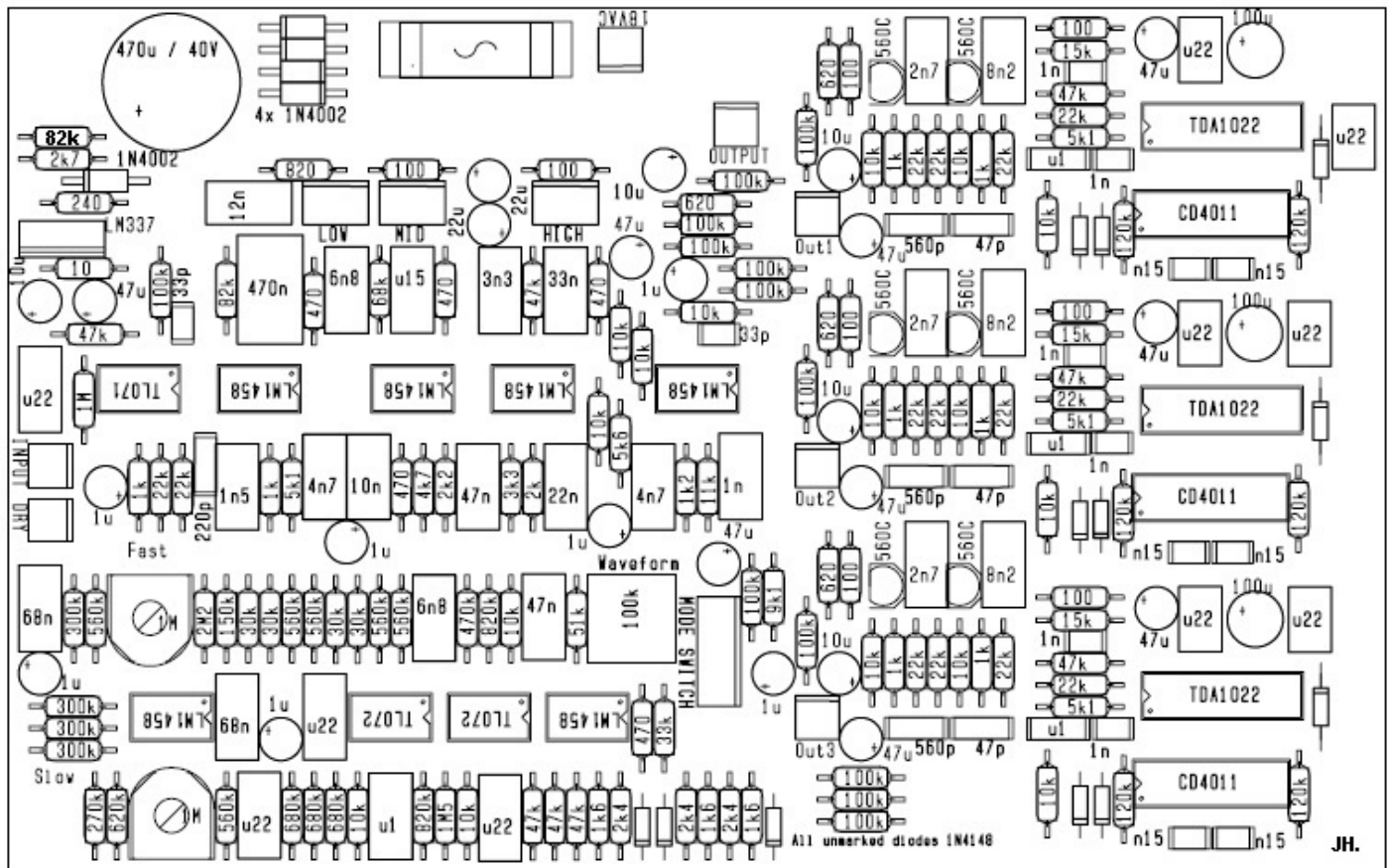
(more options to come)

The PCB board - power on-board.

It all fits onto a 160mm x 100mm board. It contains a power supply (less transformer and primary fuse). You only have to connect 18V AC from a transformer. **(And you must be qualified to handle the mains voltage - fuses, insulation, safety aspects. If in doubt, don't wire the mains voltage, but use an insulated 18V AC wallwart instead. Mains voltage can be lethal.)**

Here's a (preliminary) component overlay:

(click on the image for a full size version.)



Alternative +/-15V supply (MOTM and Synthesizers.com, etc.)

I've also included the footprints of MOTM and .COM power connectors, if you want to run the board directly from 15V, without a transformer of its own.

I've placed the footprint for the connectors beneath the (secondary) fuse, because you never need both at the same time.

When connecting to 15V DC, you **must** omit a lot of the power supply components, too.

Click on the following links for the alternative component overlay for 15V DC: [MOTM version](#), [Synthesizers.com \(.COM\) version](#).

By the way: The whole circuit only needs the -15V part of the MOTM and .COM system; the +15V pins of the connectors are unconnected.

The audio signals are centered to 0V nevertheless, of course, due to AC coupling of inputs and outputs.

Signals are weaker than your usual modular system. Input can be attenuated with a potentiometer. Output is roughly "Line level" and can go directly into a mixing desk or sound card.

Bill of Materials

Here's a [list of components](#) that are to be soldered onto the PCB.

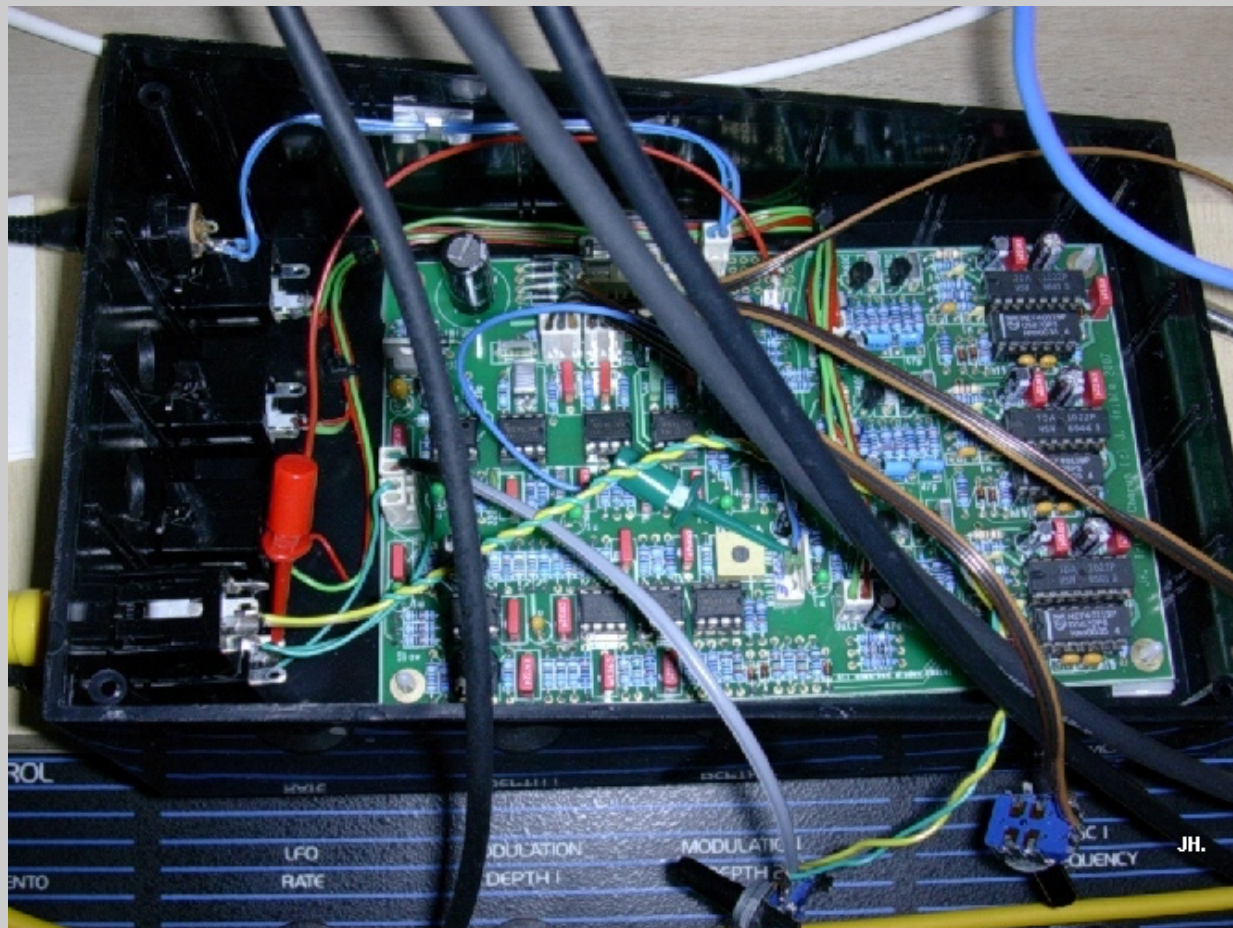
The Equalizer potentiometers are 10k linear each (3 in total).

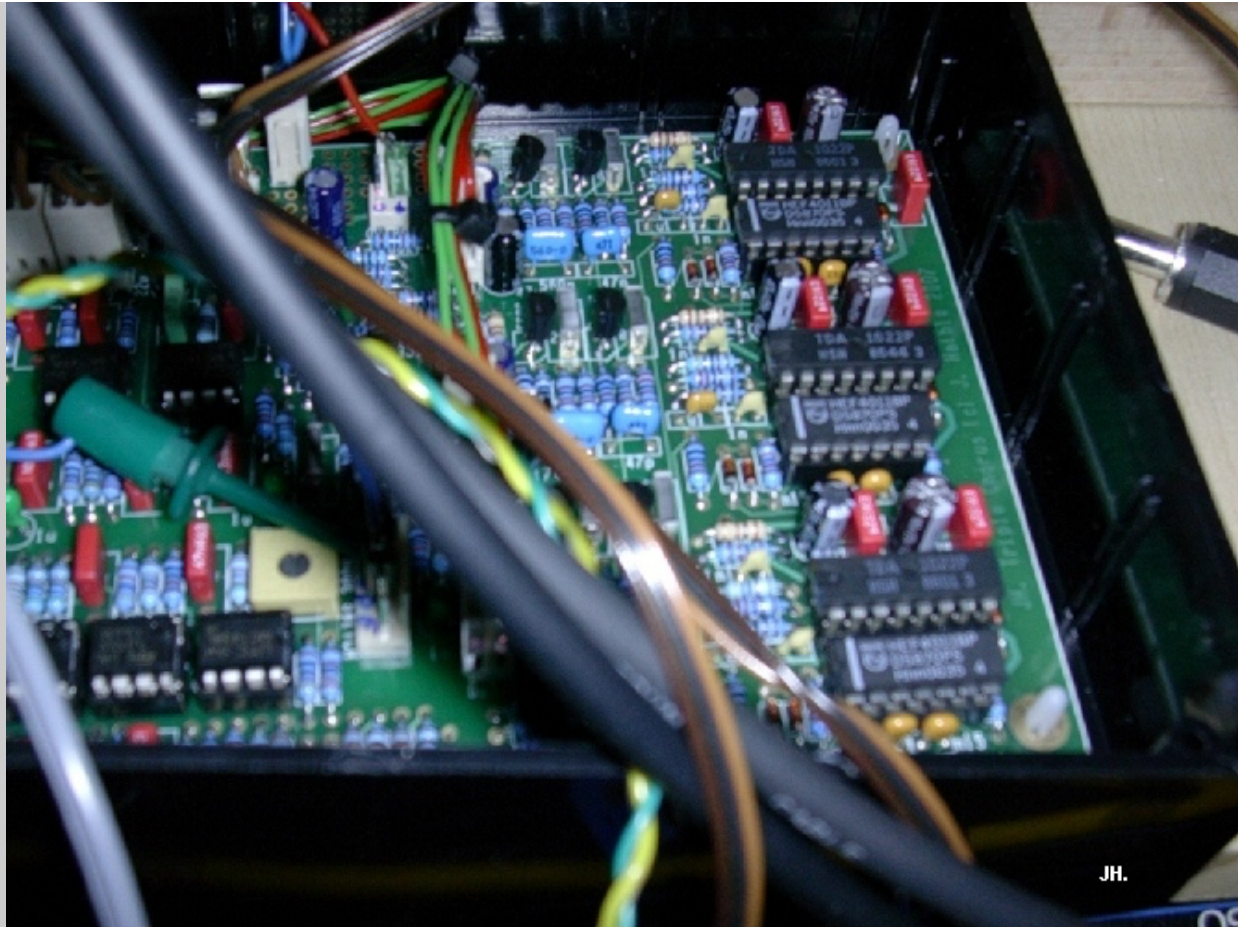
You may want to add an input level potentiometer: 10k log ... 50k log are good values.

Input / output jacks depending on what signals you want to bring out.
Mode and bypass switch can be realized with a rotary switch or with a set of toggle switches - more about this later.

Testing the Prototype

Pictures of the prototype in a small, sloped Teko enclosure; still a lot of cables and some alligator clips:





JH.

Sound Samples

Setup for demos: OB-8 (mostly set to a boring single-oscillator, unmodulated sound) -> JH Triple Chorus / String Ensemble Effect -> Reverb

(String Ensemble Effect is sometimes bypassed to show the dry sound.)

[Plain and simple string sound](#) (1.1 MB mp3)

I start with a dry sound just to show you how much the Ensemble Effect is altering it, and I switch back to the dry sound towards the end for the same reason.

[Enhancing a bell sound](#) (740 kB mp3)

I think the effect is interesting for other sounds than just mere strings, as well.

[Chorale sound \(equalized ensemble effect\)](#) (470 kB mp3)

I wouldn't sell this as a specialized choir effect device, but it's certainly an alternative to the straight Solina-like stuff. All done with the on-board 3-band Equalizer (which was inspired by the Crumar Performer).

[Stereo effect using individual outputs](#) (390 kB mp3)

Yes, we can do Stereo, with the individual outputs.

My first thought was to take the 3 individual outputs and mix them, one panned left, one center, and one right. But

then the "Ensemble" effect falls apart, and the modulation sounds overly strong. I think you can do a lot of experiments with mixing the individual outputs, and will find interesting configurations.

The above sound sample was made with a very simple setup, however:

The full mix (main output) goes to one channel, and one individual channel goes to the other. The two channels are then panned almost (but not totally) left and right on the mixing desk.

[Playing around with the EQ section](#) (1.5 MB mp3)

Just some aimless turning of the 3 potentiometers in the EQ section, while my OB-8 arpeggiates away. I obviously change the OB-8 sound to something percussive towards the end, and I also adjust the OB-8's filter envelope setting slightly for the percussive sound, but I think it's quite clear what comes from the OB-8 and what comes from the Crumar-like 3-Band EQ.

[Adjusting the waveform pot to find the "sweet spot"](#) (790 kB mp3)

What is "the Solina Sound"? I think without any modulation waveform clipping (like the sound sample starts; please also see the discussion of this topic above!), is a tad overly lush for a solina. Or maybe it isn't. During this sound sample, I chance the modulation waveform from no clipping to almost totally clipped and back, and I *think* the "Solina sound" is where the waveform is just ever so slightly clipped.

This soft clipping (the "waveform" trimpot on the board) is different from a simple attenuation of the modulation CV. It's possible to attenuate (or to boost) the modulation CV, too, by changing 3 resistors on the board, if you want to make further experiments.

Third Party Documentation (contains links to external sites; the authors of these pages are responsible for the contents.)

It's always helpful when someone else is looking at the documents I've published and double check them. Thanks to all of you who spotted the 330 ohm resistor in my BOM without quantity - it was a copy and paste error, and I fixed it in the BOM now.

Some of you are going way beyond just checking for errors, and create helpful documents of their own:

Guido Möller has set up a web site (in German) with colour coded cross-reference of components on the BOM and on the board layout.

And he even has set up a **"Warenkorb" (shopping cart) at Reichelt** with all the PCB mounted parts except the TDA1022s. He used different connectors which have the wires readily attached, which should make building the project a little easier.

Thank you, Guido!

Follow this link to his [Spezialton Solina page](#).

Mike "clickmrmike" has posted a link to his [BOM with Mouser part numbers](#). It's in spreadsheet format - thank you Mike!

I heard there's a similar page in progress with a Mouser shopping cart from another US Triple Chorus builder. I'll post a link here when it's available.

I received a [BOM for UK builders](#), mostly with Farnell part numbers, from Clive (aka Analogue) via email. Thank you!

Thanks for all who make their lists of parts available for the benefit of the community.

These lists are intended to *help* you shopping for parts - they are not guaranteed to be free of errors, and they show *one option*, among many, what parts you can buy. As with all diy projects that don't come as complete kits, you have the choice to go for different brands / quality / prices for components.

Bill & Will have started a [page documenting their progress with building the Triple Chorus](#). They also have front panel designs.

(to be continued ...)

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